

Unsupervised Brain MRI Anomaly Detection: Revisiting the StRegA Pipeline in the Era of Diffusion-based UAD

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Introduction

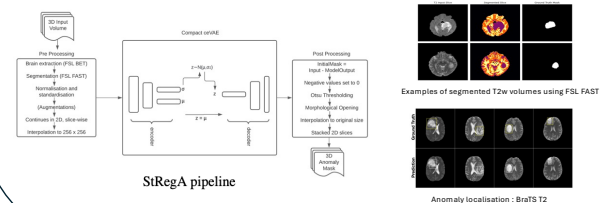
Unsupervised anomaly detection (UAD) in brain MRI aims to learn the distribution of healthy anatomy and flag out-of-distribution regions as potential lesions. StRegA (Segmentation Regularised Anomaly, 2022) is a UAD pipeline that first segments brain tissues and then uses a compact context-encoding VAE to reconstruct normal anatomy, reducing variability due to different MRI contrasts.

Its main limitations are overly smooth reconstructions and sensitivity to segmentation errors, which can hide small lesions. Newer UAD methods therefore adopt more expressive generative models and contrastive learning to obtain sharper, more reliable anomaly localisation.



Methods

StRegA is trained on MRI scans from healthy subjects in the IXI dataset. The pipeline comprises preprocessing, modelling and postprocessing. Each volume is first segmented into tissue classes with FSL-FAST, then sliced axially and resized to a fixed 2D resolution. A compact context-encoding VAE is trained on these healthy slices to reconstruct normal anatomy; at inference, reconstruction errors mark voxels that deviate from this distribution. Error maps are then refined with intensity thresholding (Otsu) and morphological filtering, and the resulting binary anomaly masks are recomposed into the original 3D image space.

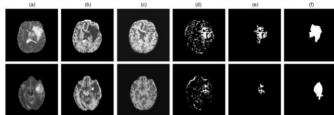


Results

StRegA was evaluated on synthetic anomalies from MOOD and on brain tumours from the BraTS dataset, and consistently outperformed VAE-based baselines such as ceVAE, GMVAE and SkipAE. On BraTS, it achieved average Dice scores of about 0.53 on T1-weighted and 0.64 on T2-weighted images, corresponding to improvements of roughly 50–80% over the best baseline ceVAE while still being trained only on anomaly-free MRIs. These results highlight the benefit of segmentation-based input simplification but also show typical UAD limitations such as under-segmentation and reduced sensitivity to very small lesions.

Model	Dice Score ($\mu \pm \sigma$)		
	MOOD Toy Data	Synthetic Anomaly Data	BraTS
SkipAE	0.517 $\pm$ 0.105	0.498 $\pm$ 0.173	0.221 $\pm$ 0.101
GMVAE	0.499 $\pm$ 0.121	0.501 $\pm$ 0.121	0.340 $\pm$ 0.167
ceVAE	0.724 $\pm$ 0.145	0.712 $\pm$ 0.132	0.580 $\pm$ 0.091
ceVAE (with StRegA pre-processing)	0.797 $\pm$ 0.089	0.783 $\pm$ 0.111	0.522 $\pm$ 0.135
StRegA	0.836 $\pm$ 0.098	0.859 $\pm$ 0.112	0.642 $\pm$ 0.101

Comparative results of StRegA with other models on T2 data.



reconstruction (b) segmented image (c) StRegA results on BraTS data set T2 images (d) input image opening threshold prediction (f) ground truth (g) morphological (h) manual otsu thresholding

Advances Beyond StRegA

Since StRegA's introduction, several modern UAD techniques have been proposed to address its limitations. Scale-Aware Contrastive Reverse Distillation (SCRD) learns multi-scale features in a student-teacher framework, improving sensitivity to both small and large lesions. Conditional diffusion models with contrastive analysis generate sharp, patient-specific healthy reconstructions, while guided diffusion models explicitly steer the reconstruction towards anatomically plausible solutions. Masked diffusion (MAD-AD) reconstructs randomly masked patches and thus learns robust healthy representations, making patch-level anomalies stand out in the reconstruction error. Normative representation learning explicitly models the variability of healthy brains to reduce false positives from benign variation, and latent Schrödinger Bridge diffusion interpolates between normal and abnormal latent distributions, providing precise anomaly localisation. Together, these methods typically outperform vanilla VAE pipelines, though their robustness on heterogeneous clinical data is still under investigation.

Conclusions

StRegA provides a strong unsupervised baseline for brain MRI anomaly detection by combining tissue segmentation with a compact generative model. Recent diffusion-based and contrastive approaches offer sharper reconstructions, richer representations and more robust anomaly localisation, suggesting clear avenues to improve upon this baseline. We plan to systematically benchmark these modern UAD methods against StRegA on IXI, MOOD and BraTS datasets to identify the most promising building blocks for future developments.

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